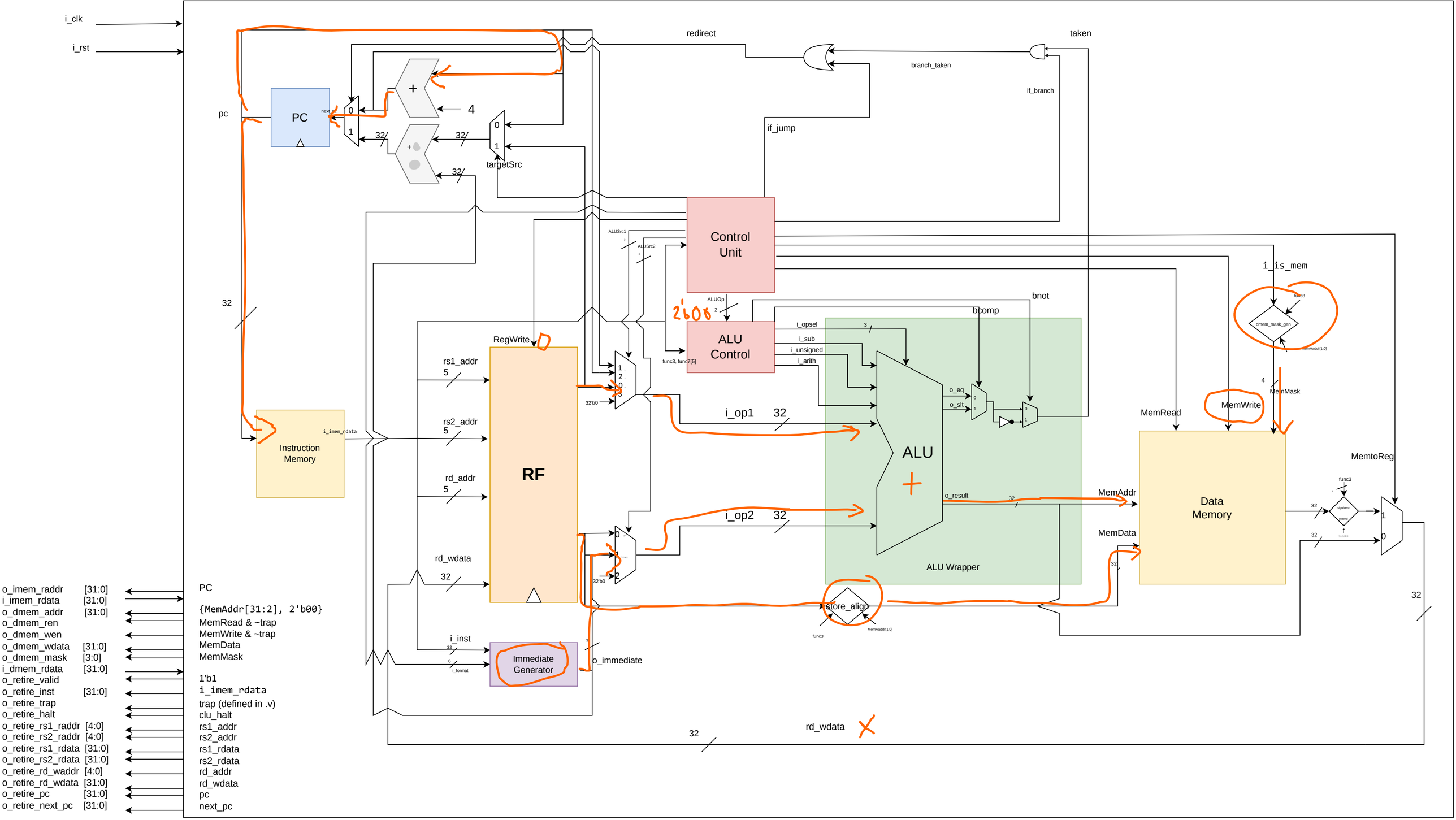


o_imem_raddr	[31:0]	PC
i_imem_rdata	[31:0]	`{MemAddr[31:2], 2'b00}`
o_dmem_addr	[31:0]	MemRead & ~trap
o_dmem_ren		MemWrite & ~trap
o_dmem_wen		MemData
o_dmem_wdata	[31:0]	MemMask
o_dmem_mask	[3:0]	
i_dmem_rdata	[31:0]	1'b1
o_retire_valid		`i_imem_rdata`
o_retire_inst	[31:0]	trap (defined in .v)
o_retire_trap		clu_halt
o_retire_halt		rs1_addr
o_retire_rs1_raddr	[4:0]	rs2_addr
o_retire_rs2_raddr	[4:0]	rs1_rdata
o_retire_rs1_rdata	[31:0]	rs2_rdata
o_retire_rs2_rdata	[31:0]	rd_addr
o_retire_rd_waddr	[4:0]	rd_wdata
o_retire_rd_wdata	[31:0]	pc
o_retire_pc	[31:0]	next_pc
o_retire_next_pc	[31:0]	